

University of Minnesota - Twin Cities

Department of Economics
4-101 Hanson Hall
1925 Fourth Street South
Minneapolis, Minnesota 55455
U.S.A.

(612) 625-6353
(612) 624-0209 FAX

Placement Director
Ellen McGrattan
(612) 625-6714 or
(612) 204-5523
erm@umn.edu
Placement Coordinator
Catherine Bach
(612) 625-6859
c-bach@umn.edu

**Curriculum Vitae
Fall 2014****SEBASTIAN DYRDA****Personal Data***Address*

4-101 Hanson Hall
1925 Fourth Street South
Minneapolis, MN 55455

Telephone Numbers

Cell: (612) 803-8530
Office: (612) 204-5497
E-mail: dyrda020@umn.edu
URL: www.dyrda.info

Citizenship: Polish (F-1 Visa)

Major Fields of Concentration

Macroeconomics, Financial Markets, Labor Economics, Public Finance

Education

<i>Degree</i>	<i>Field</i>	<i>Institution</i>	<i>Year</i>
Ph.D.	Economics	University of Minnesota (expected)	2015
M.A.	Economics	University of Minnesota	2013
M.A.	Economics	Warsaw School of Economics	2009

Dissertation

Title: "Firms Heterogeneity and Macroeconomics"

Dissertation Advisor: Professor José-Víctor Ríos-Rull

Expected Completion: Summer 2015

References

Professor José-Víctor Ríos-Rull	(612) 625-0941 vr0j@umn.edu	Department of Economics University of Minnesota 4-101 Hanson Hall
Professor V. V. Chari	(612) 626-7151 (612) 204-5518 chari002@umn.edu	1925 Fourth Street South Minneapolis, MN 55455
Professor Christopher Phelan	(612) 626-2533 cphelan@umn.edu	
Dr. Jonathan Heathcote	(612) 204-6385 heathcote@minneapolisfed.org	Research Department Federal Reserve Bank of Minneapolis 90 Hennepin Avenue South Minneapolis, MN 55480

Honors and Awards

- 2014 - 2015 *Kauffman Dissertation Fellowship*, Ewing Marion Kauffman Foundation, Kansas City, Missouri
2014 *Hella Mears Graduate Fellowship*, University of Minnesota, Minneapolis, Minnesota
Spring 2014 *Distinguished Instructor*, Department of Economics, University of Minnesota, Minneapolis, Minnesota
2013 *Graduate Research Partners Program Fellowship*, University of Minnesota, Minneapolis, Minnesota
2010 - 2011 *Distinguished Teaching Assistant*, Department of Economics, University of Minnesota, Minneapolis, Minnesota
2009 - 2010 *Leonid Hurwicz Fellowship*, University of Minnesota, Minneapolis, Minnesota

Teaching Experience

- 2012 - 2014 *Instructor*, Department of Economics, University of Minnesota, Minneapolis, Minnesota.
 Taught *Advanced Macroeconomic Policy*.
2010 - 2011 *Teaching Assistant*, Department of Economics, University of Minnesota, Minneapolis, Minnesota. Led recitation sections for *Principles of Microeconomics*.

Research Experience

- 2012 - present *Principal User*, Minnesota Supercomputing Institute, University of Minnesota, Minneapolis, Minnesota.
2011 - present *Research Analyst*, Research Department, Federal Reserve Bank of Minneapolis, Minneapolis, Minnesota.
 Research assistant for Professor José-Víctor Ríos-Rull and Dr. Francisco Buera.

Professional Experience

- 2009 *Consultant*, Board of Strategic Advisors to the Prime Minister, Warsaw, Poland
2007 - 2009 *Economist*, Institute for Structural Research, Warsaw, Poland
2007 *Intern*, World Bank, Warsaw, Poland

Papers

- “Fluctuations in Uncertainty, Efficient Borrowing Constraints, and Firm Dynamics,” job market paper
“Business Cycles and Household Formation: The Micro versus the Macro Labor Elasticity,” with José-Víctor Ríos-Rull and Greg Kaplan
“Optimal Dynamic Fiscal Policy in a Model with Uninsurable Idiosyncratic Shocks,” with Marcelo Zouain Pedroni
“Taxes, Regulations of Businesses and Evolution of Income Inequality in the US”
“Models of Government Expenditure Multipliers,” with José-Víctor Ríos-Rull
“Household Formation and Labor Market Outcomes: US vs. Europe,” with José-Víctor Ríos-Rull, in progress

Presentations

- “Fluctuations in Uncertainty, Efficient Borrowing Constraints, and Firm Dynamics,” presented at the Midwest Macroeconomics Meeting, University of Minnesota, Minneapolis, Minnesota, November 2013; Doctoral Workshop on Dynamic Macroeconomics, University of Strasbourg, Strasbourg, France, June 2014.
“Optimal Dynamic Fiscal Policy in a Model with Uninsurable Idiosyncratic Shocks,” presented at the International Conference in Computing and Finance, Vancouver, Canada, July 2013.
“Business Cycles and Household Formation: The Micro versus the Macro Labor Elasticity,” presented at the Warsaw Economic Seminars, Warsaw, Poland, January 2012.

Computer Skills

Fortran, OpenMP, MPI, Matlab, R, SAS, STATA

Languages

Polish (native), English (fluent), German (basic)

Abstracts

“Fluctuations in Uncertainty, Efficient Borrowing Constraints and Firm Dynamics,” job market paper

In this paper, I quantify the importance of microeconomic uncertainty shocks for the firm dynamics over the business cycle in an economy with frictional financial markets. To begin, I document that age rather than size is a relevant margin for the magnitude of employment volatility of firms over the cycle in the US; in particular total employment of young firms varies 2.6 times more relative to the old firms. Then, I propose a theory that, contrary to the existing studies, generates endogenously a link between firm's age and size and its ability to obtain exogenous financing. A key element of my theory is a financial friction originating from the presence of the firm's private information and long-term, efficient lending contract between a risk averse entrepreneur and financial intermediary, which manifests itself as a borrowing constraint. In times of high idiosyncratic uncertainty the financial contract calls for tightening of the borrowing constraint transmitting the initial impulse into a decline in demand for production inputs and further, including general equilibrium effects, into an economic downturn. This mechanism affects disproportionately young firms. Not only are they more constrained in borrowing but also they start smaller due to a reduced level of initial financing. A quantitative version of the model accounts for the fall of the aggregate output, employment and investment, decline of credit to GDP ratio and asymmetric employment dynamics of different groups of firms observed in the US data in recessions.

“Business Cycles and Household Formation: The Micro versus the Macro Labor Elasticity” with José-Víctor Ríos-Rull and Greg Kaplan

We provide new evidence on the cyclical behavior of household size in the United States from 1979 to 2010. During economic downturns, people live in larger households. This is mostly, but not entirely, driven by young people moving into or delaying departure from the parental home. We assess the importance of these cyclical movements for aggregate labor supply by building a model of endogenous household formation within a real business cycle structure. We use the model to measure how much more volatile are hours due mainly to two mechanisms: (i) the presence of a large group of mostly young individuals whose labor is not perfect substitute to that of older people; (ii) the facts that young people may co-reside with other people, and also change their living situation in response to aggregate conditions. These mechanisms also induce general equilibrium effects via prices that further affect the volatility of hours worked. Our exercise assumes that older people living in stable households have a Frisch elasticity that is consistent with the micro evidence that is based on such people. The inclusion of people living in unstable households yields an implied aggregate, or macro, Frisch elasticity that is at least around 72% larger than the assumed micro elasticity.

“Optimal Dynamic Fiscal Policy in a Model with Uninsurable Idiosyncratic Shocks” with Marcelo Zouain Pedroni

This paper studies the Ramsey problem in the standard infinite horizon model with heterogeneous agents and uninsurable idiosyncratic shocks. Contrary to the existing literature, we solve numerically for the optimal path of taxes along the transition between initial and final steady states. We calibrate the initial equilibrium to replicate key features of the US economy and find that the optimal levels of capital and labor income taxes should be positive and high even though the planner has undistortive lump-sum taxes as an available instrument. Capital taxes should be front-loaded hitting the upper bound for 36 initial periods then decrease to 46% in the long-run. After an initial drop labor taxes should increase towards a final level of 17%. The extra revenue is rebated via lump-sum transfers and used to lower the debt to output ratio over transition from 67% to -112.7%. To rationalize our numerical results we develop, in a simplified version of the model, an analytical characterization of the dynamic optimal tax system. We then provide two build ups to our benchmark result, quantifying the role of market incompleteness and transitory effects.

“Taxes, Regulations of Businesses and Evolution of Income Inequality in the US”

The top 10 percent income share has risen from 32.7 percent to 45.9 percent since 1980 in the US. This increase has been accompanied with a shift in its composition towards entrepreneurial income. At the same time share of pass-through entities in the distribution of the business receipts increased from 12.5 percent to 38.4 percent. In this paper I ask what was the role of changes in taxation and regulations of US businesses in accounting for these facts. To address it, I propose a dynamic, general equilibrium model with heterogeneous firms and workers in which the choice of occupation between being a worker, pass-through entity entrepreneur and co-owner of C-corporation is endogenous. Next, I study transitional effects of the policy changes since 1980 for the evolution of the distribution of businesses legal forms of organization and further I use the model to quantify policy contribution to the evolution of the income inequality in the US.